

DIGITAL POWER SUPPLY

RL78/I1A The Ultimate Power Control Microcontroller

Renesas RL78/I1A microcontrollers implements power supply control including Power Factor Correction, advanced safety and communication functions in a single chip, enabling lower cost, regulation compliance, fine dimming. RL78/I1A adopts the Renesas 16-bit RL78 CPU, which achieves both high computational performance and ultra-low power consumption, and includes 32 to 64 kilobytes of Flash memory in 20- to 38-pin packages. Safety hardware functions for IEC60730 compliance are also embedded.

Key Benefits

- Single-chip solution for power stage control.
- Variable frequency drive.
- Ultra low power consumption, high system efficiency.
- Key protection features implemented like Input Voltage (UVLO / OVLO), Output Voltage (Open load), Output short circuit.
- Supports multiple sensor & user interface controls.
- Easy integration into building automation network.
- Enables high performance intelligence power supply at low system cost.

Applications

Digital power supplies, Battery chargers etc

Device Features : RL78/I1A

- PWM High resolution with dithering (Count Clock Max 64MHz)
- Controlled in-rush current by Soft Start.
- Timer synchronous A/D capture
- Timer interlocked operation with Comparator.
- Digital Interleave PFC control function
- Enhanced Analog function
- Comparator 6ch
- Window comparator function
- PGA Amp with x4/x8/x16/x32 setting
- Hi-Z Output with Comparator TRIG

Board Specifications

- Input: 100Vac – 280Vac
- Output: 30V@1A (CCCV implementation)
- PF > 0.95
- THD < 10%
- Target efficiency > 87%
- Communication



System Block Diagram

